

The Stabilizer Formalism

Describing a code by the operators that leave it alone

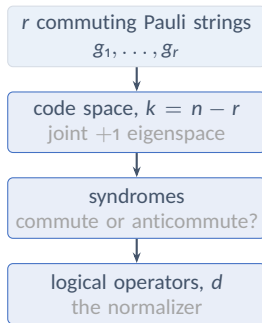
Lecture 14

Term 261

Agenda

- 1 Operators instead of kets: the Pauli group
- 2 Stabilizer groups and their code space
- 3 Syndromes, and how to measure them
- 4 Logical operators and distance
- 5 The binary picture; the perfect $[[5, 1, 3]]$ code
- 6 Clifford circuits; Gottesman-Knill

Running examples: bit-flip and Shor codes (L12),
the five-qubit code (L13).



Kets or operators?

Shor's code from L12, written two ways:

As a vector

$$|0_L\rangle = \frac{1}{2\sqrt{2}} (|000\rangle + |111\rangle)^{\otimes 3}$$

a vector of $2^9 = 512$ amplitudes (8 of them nonzero) — and $|1_L\rangle$ is another 512.

At $n = 100$ qubits a state has $2^{100} \approx 1.27 \times 10^{30}$ amplitudes. No one writes those down.

As the operators that leave it alone

$$Z_1Z_2, Z_2Z_3, Z_4Z_5, Z_5Z_6, Z_7Z_8, Z_8Z_9$$

$$X_1X_2X_3X_4X_5X_6, X_4X_5X_6X_7X_8X_9$$

Eight lines of nine letters. Every one of them maps both $|0_L\rangle$ and $|1_L\rangle$ to themselves.

At $n = 100$: about 100 strings of 100 letters.

The idea. A state is pinned down by the operators that fix it: $Z|0\rangle = |0\rangle$ and $X|+\rangle = |+\rangle$ already *define* $|0\rangle$ and $|+\rangle$. A code is pinned down the same way.

The Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Hermitian *and* unitary: $X^2 = Y^2 = Z^2 = \mathbb{I}$
- Eigenvalues ± 1 : each is an **observable**
- Two different ones **anticommute**: $XZ = -ZX$
- With $\mathbb{I}$ they span all 2×2 matrices

Up to a phase, Paulis multiply like labels; the phase $\pm i$ records the order — and decides whether two operators commute.

Multiplication table (row $\times$ column)

	$\mathbb{I}$	X	Y	Z
$\mathbb{I}$	$\mathbb{I}$	X	Y	Z
X	X	$\mathbb{I}$	iZ	$-iY$
Y	Y	$-iZ$	$\mathbb{I}$	iX
Z	Z	iY	$-iX$	$\mathbb{I}$

green: $+i$, orange: $-i$ — antisymmetric off the diagonal

Products (cyclic)

$XY = iZ, \quad YZ = iX, \quad ZX = iY$
reverse the order, flip the sign: $YX = -iZ$

The Pauli group on n qubits

Definition

$$\mathcal{G}_n = \{ \pm 1, \pm i \} \times \{ \mathbb{I}, X, Y, Z \}^{\otimes n}, \quad |\mathcal{G}_n| = 4 \cdot 4^n = 4^{n+1}$$

(checked by closure: $|\mathcal{G}_1| = 16$, $|\mathcal{G}_2| = 64$)

Notation

- $\mathbb{I}XZ = \mathbb{I} \otimes X \otimes Z$, also written X_2Z_3
- $\mathbb{I}XZ \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle) = \frac{1}{\sqrt{2}} (|010\rangle - |101\rangle)$
- **Weight** $\text{wt}(P)$: number of non- $\mathbb{I}$ positions
($\text{wt}(\mathbb{I}XZ) = 2$): how many qubits P touches

Facts we will use

- Closed under multiplication: a group
- Two elements either **commute** or **anticommute** — nothing in between
- Elements with phase ± 1 are Hermitian, square to $\mathbb{I}$, eigenvalues ± 1

Why are the phases $\{\pm 1, \pm i\}$ enough? (Every single-qubit product carries a phase from this set, and the set is closed under multiplication.)

When do two Pauli strings commute?

Swap the two strings one position at a time. A position where both are non- $\mathbb{I}$ and *different* — a **clash** — contributes a factor -1 ; every other position contributes $+1$.

Commutation rule

P and Q **commute** iff they clash in an **even** number of positions; otherwise they anticommute.

Ex. 1

X	X	I
I	Y	Z

1 clash: **anticommute**

Ex. 2

X	X	I
Z	Y	X

2 clashes: **commute**

Commute or anticommute?

- 1 $ZZ\mathbb{I}$ and $\mathbb{I}XX$ (1 clash: *anti*)
- 2 XYZ and ZYX (2 clashes: *commute*)
- 3 ZZZ and XXX (3 clashes: *anti*)

$(XXI)(Z Y X) = (XZ)(XY)(IX) = (-iY)(iZ)X = YZX$, and
the reverse order gives the same.

T251 notes, Lecture 11: $XX\mathbb{I}$ vs $\mathbb{I}YZ$ and $XX\mathbb{I}$ vs ZYX . All six pairs checked as 8×8 matrices in `L14_verify.py`.

The stabilizer of a state

An operator S **stabilizes** $|\psi\rangle$ if

$$S|\psi\rangle = |\psi\rangle,$$

i.e. $|\psi\rangle$ is a $+1$ eigenvector of S .

- Z stabilizes $|0\rangle$; $-Z$ stabilizes $|1\rangle$
- X stabilizes $|+\rangle$; $-X$ stabilizes $|-\rangle$
- If S, T stabilize $|\psi\rangle$, so do ST and S^{-1} : a **group**

$|\Phi^+\rangle$: $XX \cdot ZZ = -YY$, so its group is $\{\mathbb{I}, XX, ZZ, -YY\}$.

Takeaway: 2 qubits, 2 commuting stabilizers, 1 state: $\frac{1}{2}(\mathbb{I}+XX) \cdot \frac{1}{2}(\mathbb{I}+ZZ) = |\Phi^+\rangle\langle\Phi^+|$. Fewer stabilizers leave a whole subspace — a code.

Two commuting observables, four Bell states

	$XX = +1$	$XX = -1$
$ZZ = +1$	$ \Phi^+\rangle$ $\frac{ 00\rangle + 11\rangle}{\sqrt{2}}$	$ \Phi^-\rangle$ $\frac{ 00\rangle - 11\rangle}{\sqrt{2}}$
$ZZ = -1$	$ \Psi^+\rangle$ $\frac{ 01\rangle + 10\rangle}{\sqrt{2}}$	$ \Psi^-\rangle$ $\frac{ 01\rangle - 10\rangle}{\sqrt{2}}$
$ \Phi^+\rangle$: stabilizers XX, ZZ		$ \Phi^-\rangle$: $-XX, ZZ$

Stabilizer groups and stabilizer codes

Definition

A **stabilizer group** $\mathcal{S} \subset \mathcal{G}_n$ is **abelian** and does not contain $-\mathbb{I}$. With r independent generators, $\mathcal{S} = \langle g_1, \dots, g_r \rangle$ and $|\mathcal{S}| = 2^r$. Its **code space** is

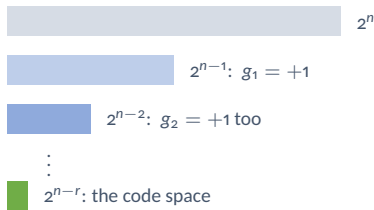
$$V_{\mathcal{S}} = \{|\psi\rangle : g|\psi\rangle = |\psi\rangle \ \forall g \in \mathcal{S}\}, \quad \dim V_{\mathcal{S}} = 2^{n-r} \Rightarrow k = n - r.$$

Why the two conditions

- $-\mathbb{I} \notin \mathcal{S}$: $-|\psi\rangle = |\psi\rangle$ forces $|\psi\rangle = 0$
- Abelian: if $g, h \in \mathcal{S}$ anticommute, $|\psi\rangle = gh|\psi\rangle = -hg|\psi\rangle = -|\psi\rangle$

Either way the code would be empty.

Why 2^{n-r} : a Pauli $g \neq \pm\mathbb{I}$ is traceless, so its ± 1 eigenspaces are equal halves; each independent generator halves what is left.



Projector onto the code space:

$$P_{\mathcal{S}} = \prod_{i=1}^r \frac{\mathbb{I} + g_i}{2} = \frac{1}{2^r} \sum_{g \in \mathcal{S}} g$$

Example 1: the bit-flip code, from its stabilizer

$$S = \langle Z_1 Z_2, Z_2 Z_3 \rangle = \{IIII, ZZII, IIZZ, ZIIZ\}: n = 3, r = 2, \text{ so } k = 1.$$

$$P_S = \frac{I + Z_1 Z_2}{2} \cdot \frac{I + Z_2 Z_3}{2} \\ = \frac{1}{4} (IIII + ZZII + IIZZ + ZIIZ)$$

All four terms are diagonal, so P_S is diagonal; read it off the table:

$$P_S = |000\rangle\langle 000| + |111\rangle\langle 111|, \quad \text{tr } P_S = 2.$$

Result

$V_S = \text{span}\{|000\rangle, |111\rangle\}$ — the code of L12, recovered from two operators.

basis state	$Z_1 Z_2$	$Z_2 Z_3$	$(P_S)_{jj}$
$ 000\rangle$	+1	+1	1
$ 001\rangle$	+1	-1	0
$ 010\rangle$	-1	-1	0
$ 011\rangle$	-1	+1	0
$ 100\rangle$	-1	+1	0
$ 101\rangle$	-1	-1	0
$ 110\rangle$	+1	-1	0
$ 111\rangle$	+1	+1	1

the other six states pair up into the three other ± 1 patterns — the error spaces

Errors flip stabilizer eigenvalues

Let $|\psi\rangle \in V_S$ and let a Pauli error E act. For each generator g_i :

$$g_i (E |\psi\rangle) = \pm E g_i |\psi\rangle = \pm E |\psi\rangle \quad \begin{cases} + & E \text{ commutes with } g_i \Rightarrow s_i = 0 \\ - & E \text{ anticommutes with } g_i \Rightarrow s_i = 1 \end{cases}$$

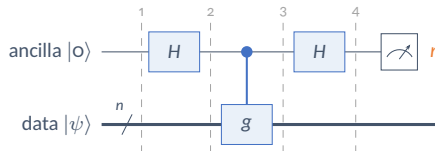
The **syndrome** $\mathbf{s} = (s_1, \dots, s_r)$ is read off by counting clashes — no state vectors needed.

E	clashes w/ ZZI, IZZ	$(Z_1 Z_2, Z_2 Z_3)$	recovery
none	0, 0	(+1, +1)	—
X_1	1, 0	(-1, +1)	X_1
X_2	1, 1	(-1, -1)	X_2
X_3	0, 1	(+1, -1)	X_3
Y_1	1, 0	(-1, +1)	looks like X_1
Z_1	0, 0	(+1, +1)	invisible

- Same table as L12, now *derived*
- $\mathbf{s}$ depends on E only, never on α, β : checked with $\alpha |000\rangle + \beta |111\rangle$ in a statevector
- An error that commutes with every g_i gives $\mathbf{s} = \mathbf{0}$: it is either harmless or a **logical** error

Measuring a stabilizer: the Hadamard test

We need the eigenvalue of g (a Pauli string, $g^2 = \mathbb{I}$) *without* measuring the data qubits.



- 1 $|0\rangle |\psi\rangle$
- 2 $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |\psi\rangle$
- 3 $\frac{1}{\sqrt{2}}(|0\rangle |\psi\rangle + |1\rangle g |\psi\rangle)$
- 4 $|0\rangle \frac{\mathbb{I}+g}{2} |\psi\rangle + |1\rangle \frac{\mathbb{I}-g}{2} |\psi\rangle$

Reading the outcome ($\frac{\mathbb{I} \pm g}{2}$ are projectors because $g^2 = \mathbb{I}$)

$$\Pr(m = 0) = \langle \psi | \frac{\mathbb{I} + g}{2} | \psi \rangle = \frac{1 + \langle g \rangle}{2}$$

and the data is left in $\frac{\mathbb{I} \pm g}{2} |\psi\rangle$, renormalized: a projective measurement of g .

On a code state, after an error E

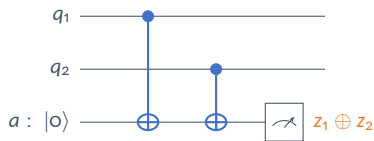
$E |\psi\rangle$ is an eigenvector of g , so $m = s_i$ **with certainty** and the data is **unchanged**: the measurement reveals the syndrome and nothing about α, β .

Same circuit as the Hadamard test of L9. Probabilities and post-measurement states checked in L14_verify.py.

The CNOT shortcut for Z-type and X-type checks

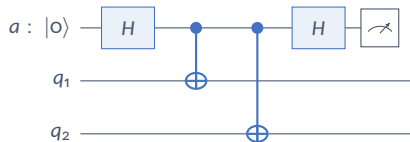
For Pauli strings the controlled-g splits into two-qubit gates, one per letter:

Z-check $Z_1 Z_2$: CNOTs into the ancilla



The ancilla ends in $|z_1 \oplus z_2\rangle$: the parity, and only the parity. This is the Hadamard test, since $H_a CZ H_a = \text{CNOT into } a$.

X-check $X_1 X_2$: CNOTs out of the ancilla



Controlled- $(X_1 X_2)$ is literally two CNOTs from the ancilla. Measuring $X = H$ then measure Z .

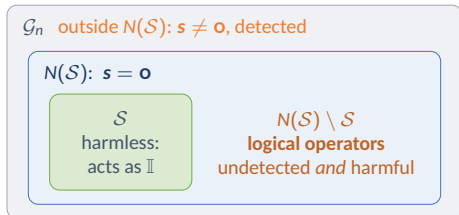
For $\alpha |000\rangle + \beta |111\rangle$, what state is the $Z_1 Z_2$ ancilla in after the two CNOTs? ($|0\rangle$, untouched by α, β : the ancilla never becomes entangled with the logical information.)

L12's syndrome circuit is two copies of the left one. A mixed check ($XZZXII$) uses one controlled-X or -Z per letter.

Logical operators and distance

The **normalizer** $N(\mathcal{S})$: Paulis that commute with every element of $\mathcal{S}$.

If $L \in N(\mathcal{S})$ and $|\psi\rangle \in V_{\mathcal{S}}$: $gL|\psi\rangle = Lg|\psi\rangle = L|\psi\rangle$, so L keeps us *inside* the code space.



Distance of a stabilizer code

$$d = \min \{ \text{wt}(L) : L \in N(\mathcal{S}) \setminus \mathcal{S} \}$$

the lightest error that slips through and changes the logical state.

Bit-flip code

- $\bar{X} = XXX$: $|000\rangle \leftrightarrow |111\rangle$
- $\bar{Z} = Z_1$ (or $ZZZ = Z_1 \cdot \mathbb{I}ZZ$): $|1_L\rangle \rightarrow -|1_L\rangle$
- Brute force over all 64 Paulis: the lightest logicals are Z_1, Z_2, Z_3 , so $d = 1$

$\Rightarrow$ a $[[3, 1, 1]]$ code, as in L13: distance 3 only against bit flips.

Example 2: Shor's code in eight lines

	1	2	3	4	5	6	7	8	9
g_1	Z	Z							
g_2		Z	Z						
g_3				Z	Z				
g_4					Z	Z			
g_5							Z	Z	
g_6								Z	Z
g_7	X	X	X	X	X	X			
g_8				X	X	X	X	X	X
	block 1			block 2			block 3		

all 28 pairs commute; rank 8 $\Rightarrow k = 9 - 8 = 1$;
 $\text{tr } P_S = 2$

Logical operators — found by applying candidates to the 512-dim codewords:

operator	action on code	role
$Z_1 Z_4 Z_7 \ (\equiv Z^{\otimes 9})$	$ 0_L\rangle \leftrightarrow 1_L\rangle$	$\bar{X}$
$X_1 X_2 X_3 \ (\equiv X^{\otimes 9})$	$ 1_L\rangle \rightarrow - 1_L\rangle$	$\bar{Z}$
Z_1, X_1	leave the code space	detectable

- Letters swap (the outer layer is a phase-flip code): names follow the *action*, not the letters
- $d = 3$: brute force over all $4^9 = 262,144$ Paulis $\Rightarrow [[9, 1, 3]]$
- Degenerate:** Z_1, Z_2 share a syndrome but act identically ($Z_1 Z_2 \in \mathcal{S}$); 27 single-qubit errors, 21 syndromes, all corrected

Generators as in the T251 notes, Lecture 11, with qubits numbered 1–9.

The binary picture: Paulis as bit vectors

Drop the phase and record which positions carry an X-part and which a Z-part:

$$P \mapsto (\mathbf{x} | \mathbf{z}) \in \mathbb{F}_2^{2n}, \quad \mathbb{I} \mapsto (0|0), \quad X \mapsto (1|0), \quad Z \mapsto (0|1), \quad Y \mapsto (1|1).$$

Multiplication $\rightarrow$ addition mod 2

$$\begin{array}{rcl} XZ\mathbb{I} & \mapsto & (100 | 010) \\ ZX\mathbb{I} & \mapsto & (010 | 100) \\ \hline XZ\mathbb{I} \cdot ZX\mathbb{I} = YY\mathbb{I} & \mapsto & (110 | 110) \end{array}$$

Check matrix: generators as rows, $r \times 2n$.

Bit-flip code, $\langle Z_1 Z_2, Z_2 Z_3 \rangle$:

$$H = \left[\begin{array}{ccc|cc} 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right]$$

Commutation $\rightarrow$ symplectic product

$$P, Q \text{ commute} \iff \mathbf{x} \cdot \mathbf{z}' + \mathbf{z} \cdot \mathbf{x}' = 0 \pmod{2}$$

(checked against the matrices on 2000 random pairs)

- generators independent $\iff$ rows independent over $\mathbb{F}_2$ (rank r)
- abelian $\iff H\Lambda H^T = 0$, $\Lambda = \begin{bmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{bmatrix}$
- syndrome of $E = (\mathbf{x} | \mathbf{z})$: $\mathbf{s} = H\Lambda (\mathbf{x} | \mathbf{z})^T$

Takeaway: A stabilizer code is a binary matrix whose rows are symplectically orthogonal. CSS codes (L15), the surface code and qLDPC codes (L16) are all ways of choosing H .

Example 3: the five-qubit code $[[5, 1, 3]]$

Each generator is a cyclic shift of the one above (T251 notes, L14: the “perfect code”):

	1	2	3	4	5
g_1	X	Z	Z	X	$\mathbb{I}$
g_2	$\mathbb{I}$	X	Z	Z	X
g_3	X	$\mathbb{I}$	X	Z	Z
g_4	Z	X	$\mathbb{I}$	X	Z
$\bar{X}$	X	X	X	X	X
$\bar{Z}$	Z	Z	Z	Z	Z

$$H = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$$

- All 6 pairs of generators clash in exactly 2 positions: they commute
- $\text{rank}_{\mathbb{F}_2} H = 4 \Rightarrow k = 5 - 4 = 1$ (the fifth shift $ZZX\mathbb{I}X$ is $g_1 g_2 g_3 g_4$, not new)
- $\bar{X}, \bar{Z}$ commute with every g_i , are not in $\mathcal{S}$, and anticommute with each other (5 clashes)
- $d = 3$: $\bar{X}g_1 = -\mathbb{I}YY\mathbb{I}X$ has weight 3; no lighter logical among all $4^5 = 1024$ Paulis

Result

$[[5, 1, 3]]$: any single-qubit error corrected with 5 qubits instead of Shor's 9.

Gottesman, PRA 54, 1862; Laflamme et al., PRL 77, 198; Bennett et al., PRA 54, 3824 (all 1996); Nielsen & Chuang §10.5.6.

Why $[[5, 1, 3]]$ is perfect: every syndrome is used once

Syndrome bit $s_i = 1$ iff the error clashes with g_i an odd number of times (here: once).

	qubit 1			qubit 2			qubit 3			qubit 4			qubit 5		
	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
g_1	0	1	1	1	1	0	1	1	0	0	1	1	0	0	0
g_2	0	0	0	0	1	1	1	1	0	1	1	0	0	1	1
g_3	0	1	1	0	0	0	0	1	1	1	1	0	1	1	0
g_4	1	1	0	0	1	1	0	0	0	0	1	1	1	1	0
as a number	1	11	10	8	13	5	12	14	2	6	15	9	3	7	4

- The 15 columns are $1, 2, \dots, 15$, each exactly once: all distinct, all nonzero
- With “no error” = 0000: $1 + 3 \cdot 5 = 16 = 2^4$ error classes for 2^4 syndromes

Quantum Hamming bound (L13)

$$k = 1, t = 1: 2(1 + 3n) \leq 2^n$$

$$\text{at } n = 5: 32 \leq 32 - \text{equality}$$

Worked decoding: a Y error on qubit 3

Start from $|\psi\rangle = \alpha |0_L\rangle + \beta |1_L\rangle$ (a 32-dimensional vector; $|0_L\rangle$ has 16 nonzero amplitudes) and apply $E = Y_3 = \text{IIIYIII}$.

1. Measure the four generators. Only position 3 matters:

	g_1	g_2	g_3	g_4
letter at position 3	Z	Z	X	I
clash with Y?	yes	yes	yes	no
eigenvalue	-1	-1	-1	+1
syndrome bit	1	1	1	0

2. Look up 1110 (= 14) in the table: the only match is Y_3 .

3. Apply Y_3 again: $Y_3 Y_3 = I$.

Statevector check, 32 amplitudes

$\alpha = \cos 0.3$, $\beta = e^{0.7i} \sin 0.3$. On $Y_3 |\psi\rangle$ the generators give exactly $(-1, -1, -1, +1)$; fidelity with $|\psi\rangle$ after recovery: 1.000000.

A coherent error

$$e^{-0.2iY_3} = \cos 0.2 I - i \sin 0.2 Y_3:$$

- 0000 w.p. $\cos^2 0.2 = 0.9605$: do nothing
- 1110 w.p. $\sin^2 0.2 = 0.0395$: apply Y_3

Both branches return $|\psi\rangle$ exactly: the syndrome measurement turned a rotation into a discrete error.

The Clifford group, and the Gottesman-Knill theorem

Clifford U : $UPU^\dagger \in \mathcal{G}_n$ for every Pauli P .
Generated by H, S, CNOT .

P	X	Y	Z
$HPH^\dagger$	Z	$-Y$	X
$SPS^\dagger$	Y	$-X$	Z
$\text{CNOT}_{1 \rightarrow 2}$			
$X\mathbb{I} \rightarrow XX$	$\mathbb{I}X \rightarrow \mathbb{I}X$		
$Z\mathbb{I} \rightarrow Z\mathbb{I}$	$\mathbb{I}Z \rightarrow ZZ$		

CNOT copies X forward (control $\rightarrow$ target) and Z backward (target $\rightarrow$ control).

Not Clifford: $TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$ is not a Pauli.

If $g|\psi\rangle = |\psi\rangle$ then $(UgU^\dagger)U|\psi\rangle = U|\psi\rangle$: a **Clifford gate just rewrites the generators**.

Gottesman-Knill theorem

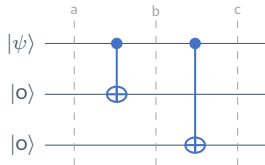
Stabilizer states, Clifford gates and Pauli measurements can be simulated efficiently on a classical computer: track n generators of $2n$ bits, not 2^n amplitudes.

- **Useful:** codes, encoders, syndrome circuits are simulated this way
- **A limit:** Cliffords alone give no speed-up; universality needs a non-Clifford gate such as T

Gottesman, quant-ph/9807006 (1998); Aaronson & Gottesman, PRA 70, 052328 (2004); N&C §10.5.2–10.5.4.

Worked: the bit-flip encoder, tracked by its generators

Encode $|\psi\rangle |0\rangle |0\rangle$ with two CNOTs (L12). Instead of following amplitudes, push the generators through the table.



Before encoding, qubits 2 and 3 are fixed by Z_2 and Z_3 ; the data qubit carries the logical X and Z .

	(a) start	(b) after $\text{CNOT}_{1 \rightarrow 2}$	(c) after $\text{CNOT}_{1 \rightarrow 3}$
stabilizer	$II\bar{I}$	$ZZ\bar{I}$	$ZZ\bar{I}$
stabilizer	$III\bar{Z}$	$III\bar{Z}$	$ZII\bar{Z}$
logical $\bar{X}$	$XIII$	$XX\bar{I}$	XXX
logical $\bar{Z}$	$ZIII$	$ZIII$	$ZIII$

$\langle ZZ\bar{I}, ZII\bar{Z} \rangle = \langle ZZ\bar{I}, III\bar{Z} \rangle$ since $ZZ\bar{I} \cdot ZII\bar{Z} = III\bar{Z}$: the bit-flip code, with $\bar{X} = XXX$ and $\bar{Z} = Z_1$ — exactly what we found from the normalizer.

Takeaway: Four Pauli strings, each rewritten by a table look-up at every gate: that is the whole Gottesman-Knill simulation — and a quick way to check any encoder.

Summary: a stabilizer code in four lines

- 1 **Generators:** r independent, commuting Pauli strings, $-\mathbb{I} \notin \mathcal{S}$
- 2 **Size:** $k = n - r$ logical qubits;
 $P_{\mathcal{S}} = \prod_i \frac{\mathbb{I} + g_i}{2}$
- 3 **Syndromes:** $s_i = 1$ iff E anticommutes with g_i — count clashes
- 4 **Logicals and distance:** $N(\mathcal{S}) \setminus \mathcal{S}$; $d =$ lightest logical

Measuring g_i costs one ancilla and one two-qubit gate per letter; Clifford circuits only rewrite generators.

	bit-flip	Shor	five-qubit
n, r	3, 2	9, 8	5, 4
$k = n - r$	1	1	1
$\bar{X}$	XXX	$Z_1 Z_4 Z_7$	XXXXX
$\bar{Z}$	Z_1	$X_1 X_2 X_3$	ZZZZZ
d	1	3	3
code	[[3, 1, 1]]	[[9, 1, 3]]	[[5, 1, 3]]

L15: choose every generator all-X or all-Z, and the check matrix splits into two classical parity-check matrices — CSS codes and the Steane code.

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Every commutation, syndrome table, logical operator, projector and distance on these slides was computed in `scripts/L14_verify.py` (numpy statevectors and $\mathbb{F}_2$ linear algebra).